

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2025

CS3001-1 – COMPUTER NETWORK AND COMMUNICATION

(For CS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. In a fully connected mesh topology with six computers, we need _____ links.
A) 6
B) 15
C) 36
D) 30
2. To deliver a message to the correct application program running on a host, the _____ address must be consulted.
A) port
B) IP
C) physical
D) Host
3. _____ is a type of transmission impairment in which the signal loses strength due to the resistance of the transmission medium.
A) Attenuation
B) Distortion
C) Decibel
D) Noise
4. _____ conversion involves three techniques: line coding, block coding, and scrambling.
A) Analog-to-digital
B) Digital-to-analog
C) Analog-to-analog
D) Digital-to-digital
5. Which transmission media provides the highest transmission speed in a network?
A) coaxial cable
B) optical fiber
C) twisted pair cable
D) electrical cable
6. Baseband transmission of a digital signal is possible only if we have a _____ channel.
A) Bandpass
B) Low rate
C) High rate
D) Low-pass
7. The _____ sublayer is responsible for the operation of the CSMA/CD access method and framing.
A) LLC
B) MII
C) MAC
D) IP
8. When a collection of various computers appears as a single coherent system to its clients, what is this called?
A) Mail system
B) Networking system
C) Computer network
D) Distributed system
9. During a Zoom meeting, users complain of delayed audio while video frames still arrive, but out of sync. Which QoS metric is most affected?
A) Throughput
B) Bandwidth
C) Jitter
D) Packet loss
10. The main header of an IPv6 protocol is _____ bytes.
A) 20
B) 40
C) 60
D) 80
11. In virtual circuit network each packet contains _____
A) full source and destination address
B) a short VC number
C) only source address
D) only destination address



12. During error reporting, ICMP always reports error messages to _____.
 A) Destination B) Source
 C) Next router D) Previous router
13. Which sequence of procedure is correct to build a routing table in link state routing?
 A) Creating LSP → flooding → shortest path tree → calculating routing table
 B) Shortest path tree → flooding → creating LSP → calculating routing table
 C) Shortest path tree → flooding → calculating routing table - creating LSP
 D) Creating LSP → flooding → calculating routing table → shortest path tree
14. Which algorithm is implemented based on link state routing?
 A) OSLR B) BGP
 C) RIP D) OSPF
15. In subnetting, the number of addresses in each subnet must _____.
 A) be a multiple of 128 B) be divisible by 128
 C) be a power of 2 D) Be a multiple of 4
16. What is start frame delimiter (SFD) in ethernet frame?
 A) 10101010 B) 11111111
 C) 00000000 D) 10101011
17. In HTTP _____ server is a computer that keeps copies of responses to recent requests.
 A) a regular B) a proxy
 C) an auxiliary D) a remote
18. Which one of the following allows a user at one site to establish a connection to another site and then pass keystrokes from local host to remote host?
 A) HTTP B) FTP
 C) Telnet D) TCP
19. DNS server translates *www.amazon.com* into an IP address. In this interaction, which statement is correct?
 A) DNS is an application layer protocol that usually uses UDP
 B) DNS is a transport layer protocol using TCP only
 C) DNS is a link layer protocol
 D) DNS acts like a client in every case
20. A socket address is a combination of _____.
 A) MAC address and Logical Address
 B) MAC address and Port Number
 C) Logical Address and Port Number
 D) User-specific address and Logical Address

PART - B: DESCRIPTIVE ANSWER QUESTIONS

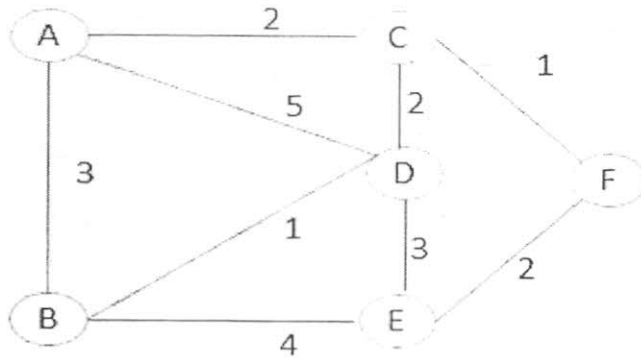
Unit – I

	Marks	BT*	CO*	PO*
1. a) Explain the four basic network topologies with advantages and disadvantages.	8	L2	1	1
b) Explain the working principle of twisted pair cable in detail.	5	L2	1	1
c) What are the propagation time and the transmission time for a 2.5kbyte message (an e-mail) if the bandwidth of the network is 1Gbps? Assume that the distance between the sender and the receiver is 12,000 km and that light travels at 2.4×10^8 m/s.	3	L3	1	2
2. a) With neat diagram explain briefly the ISO-OSI Reference Layer model in network.	10	L2	1	1
b) Explain the Ethernet frame format with diagram.				
3. a) Solve using Cyclic Redundancy Check both encoding and decoding. Dataword: 1001, Divisor is 1011.	6	L3	2	2
b) What is CSMA/CD. Describe the flow diagram of CSMA/CD.	6	L2	2	1
c) Explain the different types of transmission impairment causes.	4	L2	2	1

Unit – II

4. a) Define Virtual Circuit Network (VCN). What are the 3 phases of VCN? Explain the virtual circuit packet switching with neat diagram.	8	L2	2	1
b) Discuss the Classful addressing with a neat diagram.	8	L2	3	1

5. a) Explain the IPv4 addressing header format with diagram 8 L2 3 1
 b) Solve the Link state algorithm by using following diagram with the algorithms.



6. a) Different criteria used to measure the network performance. 8 L3 4 2
 b) A university has been granted a block of addresses with the beginning address 172.24.16.0/24. The university needs to have 3 subblocks of addresses to use in its three subnets: one subblock of 15 addresses for labs, one subblock of 62 addresses for faculties, and one subblock of 125 addresses for students. (we assign addresses to subblocks, starting with the largest and ending with the smallest one)
 i. Find out first and last addresses.
 ii. Design the subblocks. 4 L2 3 1
 c) Describe the Leaky Bucket and Token Bucket algorithm. 6 L3 3 2
 6 L2 4 1

Unit – III

7. a) Explain the TCP header format with neat diagram. 8 L2 5 1
 b) Discuss the features of HTTP and also discuss how HTTP works. 8 L2 5 1
 8. a) Describe the connectionless and connection-oriented services provided by Transport layer. 8 L2 5 1
 b) Explain the working of electronic mail and describe the Mail Transfer Phases 8 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
